Lean Management (and Lego Simulation)

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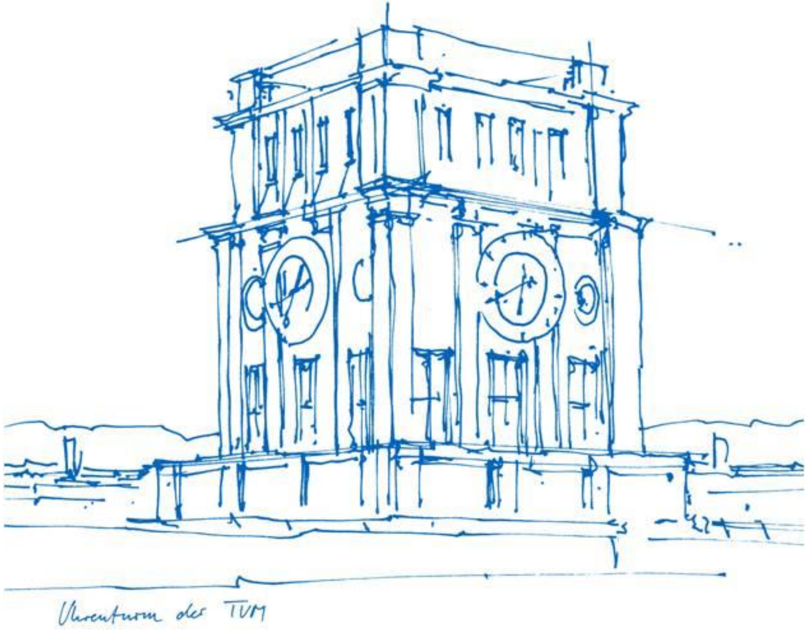
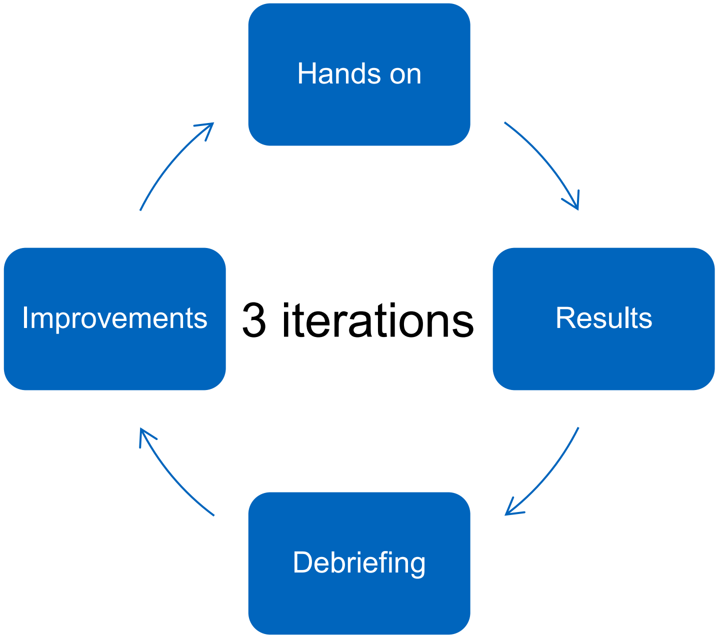


Lean Lego Simulation

Simulation



Your main product

44 bricks
per glider

1 glider sold
= $ 10,000

1 brick = $ 100

“Sly Slider” introduced in 2009

Margin per glider = $ 5,600

“Icky Ice Glider” introduced in 2011

Margin per glider = $ 5,600

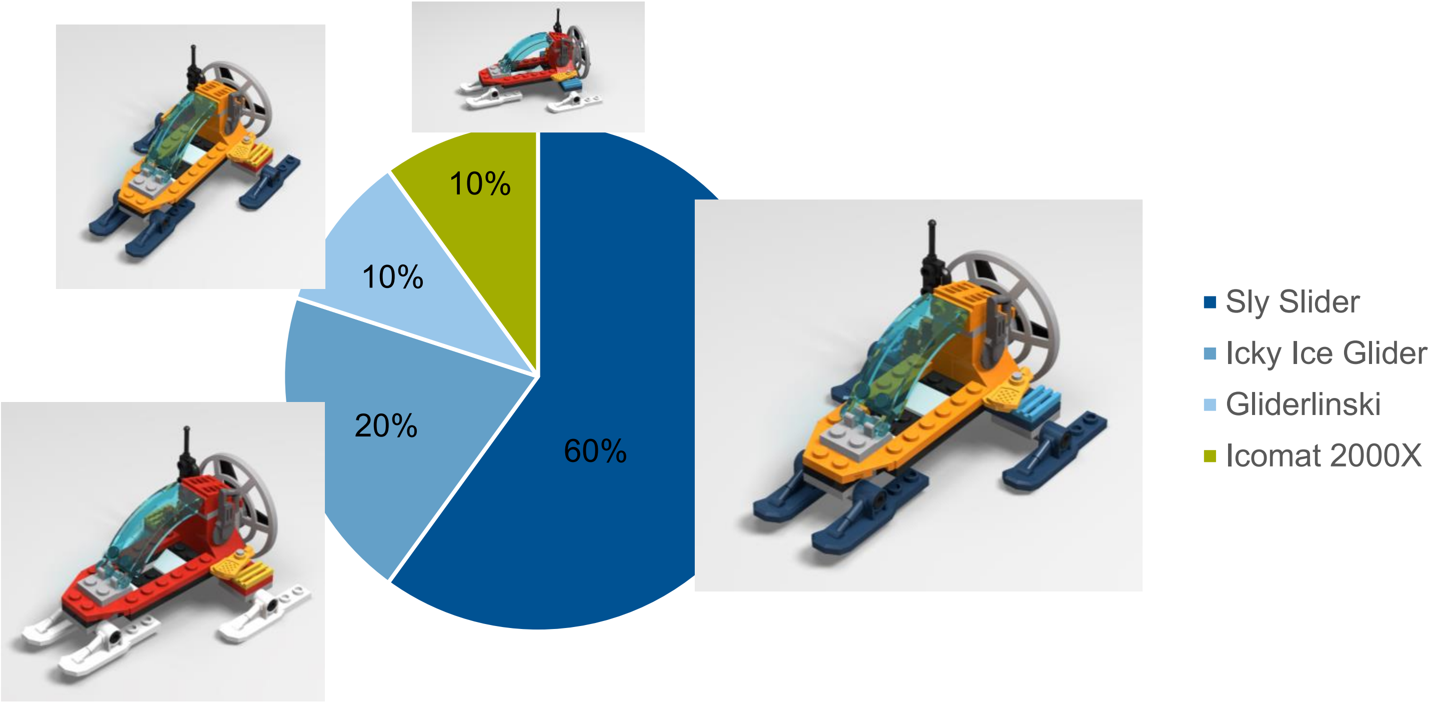
“Gliderlinski” introduced 2013

Margin per glider = $ 5,600

"Icomat 2000X" introduced 2016

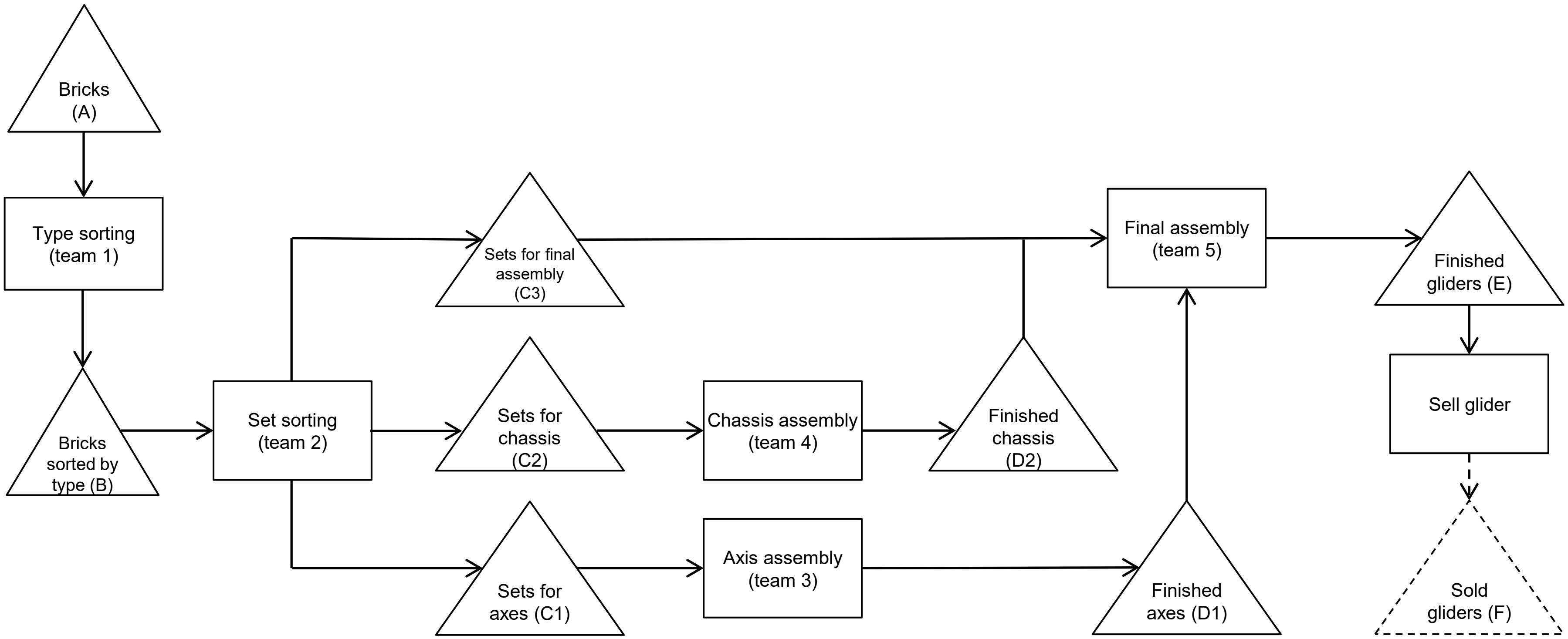
Margin per glider = $ 5,600

Past sales (2018)



Iteration 1

Iteration 1: Conventional Production Process



Sequence

Taking inventory (assess status quo)

1st production run: 3 minutes production (1 minute corresponds to 1 week in real)

Taking inventory

2nd production run: 3 minutes production (1 minute corresponds to 1 week in real)

Taking inventory

3rd production run: 3 minutes production (1 minute corresponds to 1 week in real)

Taking inventory

4th production run: 3 minutes production (1 minute corresponds to 1 week in real)

Taking inventory

Timer (3 minutes)

Minute 1 Minute 2 Minute 3

Start!

please stop producing!

Debrief: Iteration 1

Did you observe *muda*?

Muda or the 7 Wastes of Manufacturing

- Overproduction ahead of demand
- Unnecessary transport of materials
- Over-processing of parts
- Inventories above the minimum
- Unnecessary movements by employees during their work
- Production of defective parts
- Waiting

Push versus Pull

Push

Pull

Kanban (queue limiter)

Kanban

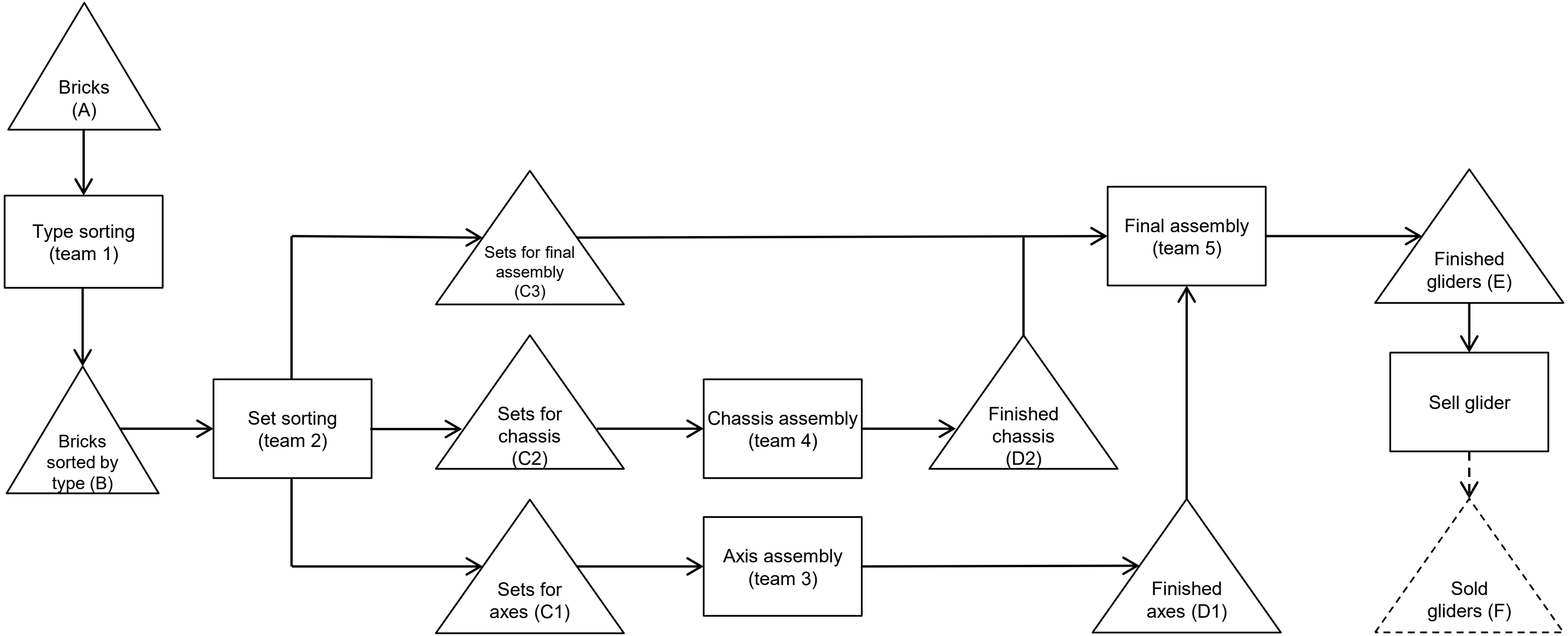
- Demand signaling
- Queue limiter (=work in progress limiter)
- Visual control
- Internal supermarket
- Self-directing

Kanban in practice

https://www.wuerth-industrie.com/web/de/wuerthindustrie/cteile_management/kanban/kanbansysteme_kanbanprinzip/systeme_prinzip_kanban.php

Iteration 2: Pull with Kanban

Iteration 2: Pull with Kanban



Sequence

Taking inventory (assess status quo)

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Taking inventory

2nd production run: 3 minutes production (1 minute corresponds to 1 week in real)

Taking inventory

3rd production run: 3 minutes production (1 minute corresponds to 1 week in real)

Taking inventory

4th production run: 3 minutes production (1 minute corresponds to 1 week in real)

Taking inventory

Timer (3 minutes)

Minute 1

Minute 2

Minute 3

Start!

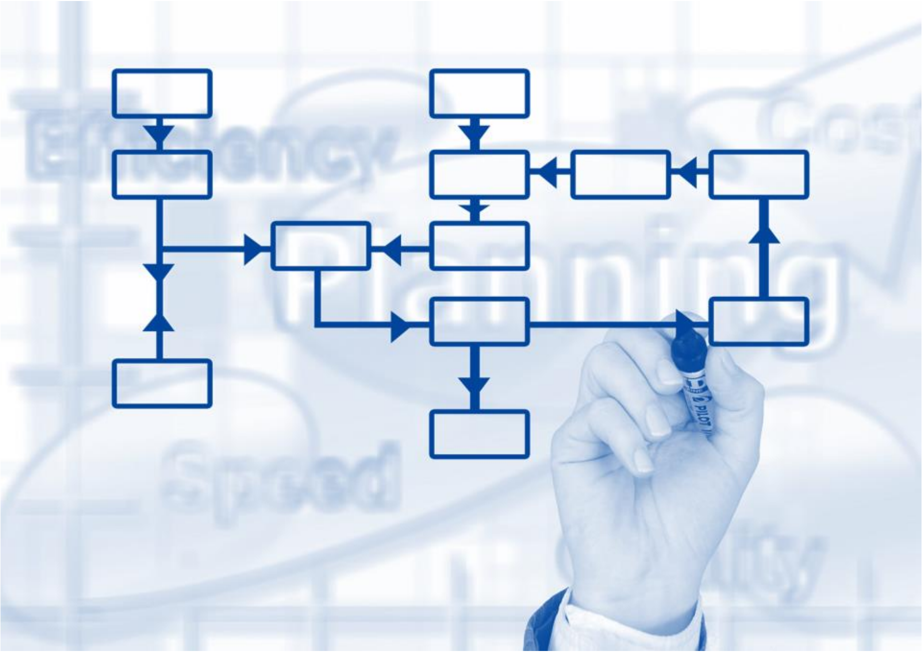
please stop producing!

Debrief: Iteration 2

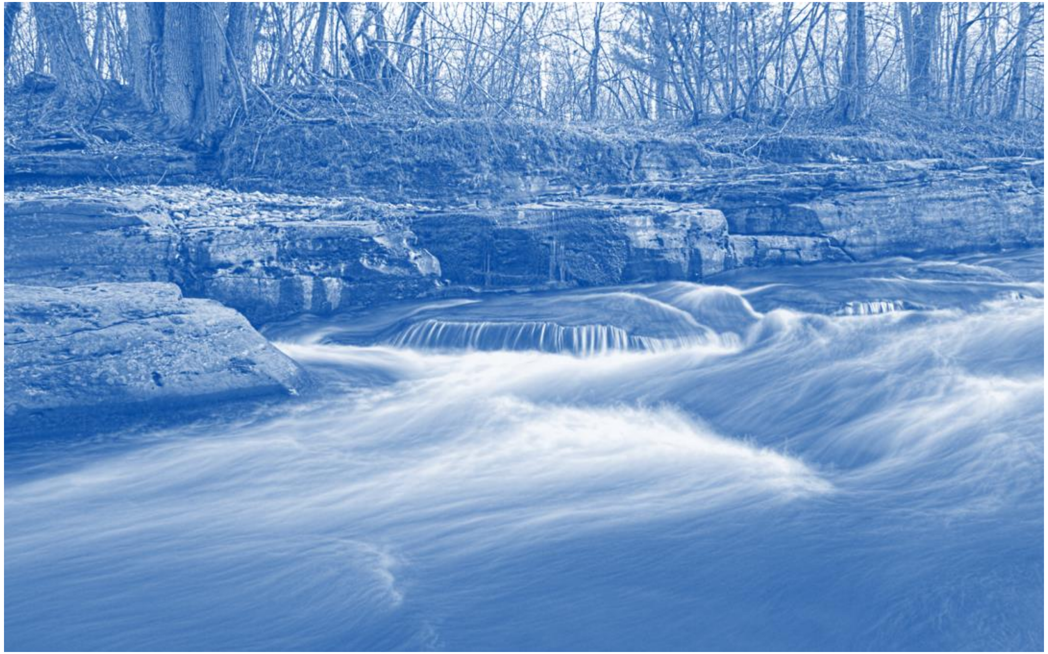
What went wrong?

Value

Value Stream



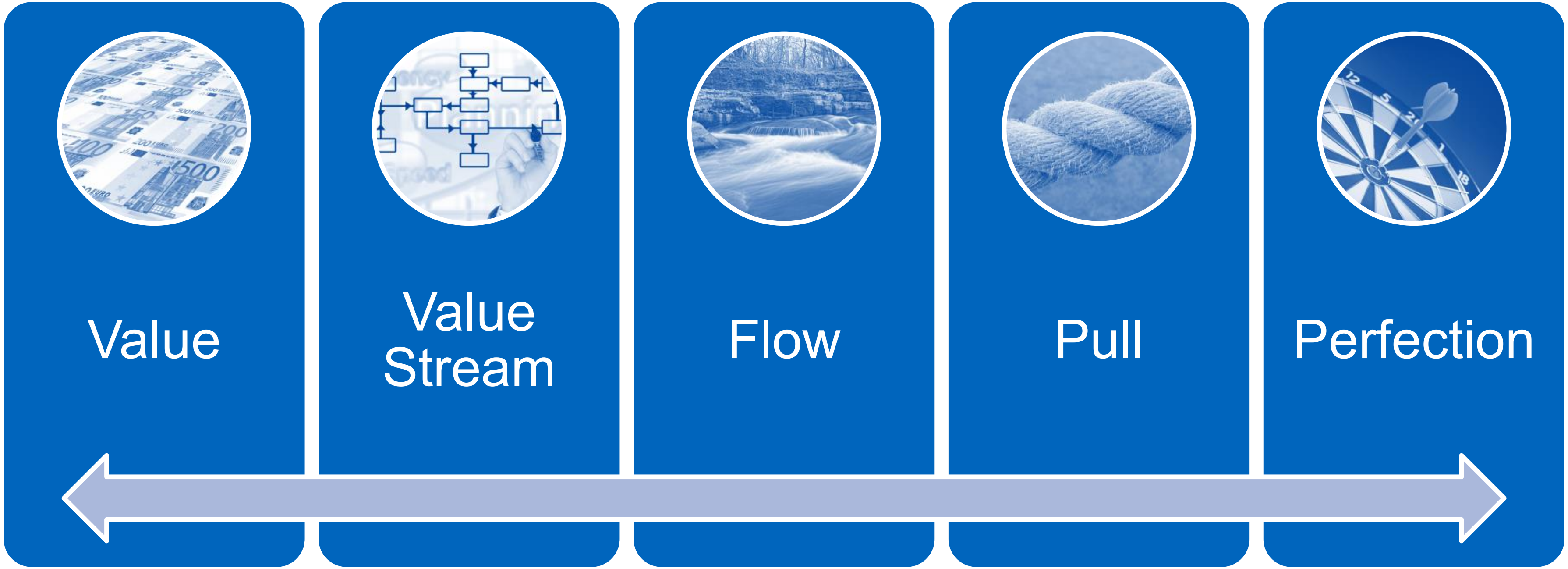
Flow



Pull

Perfection

Five Elements of Lean Management



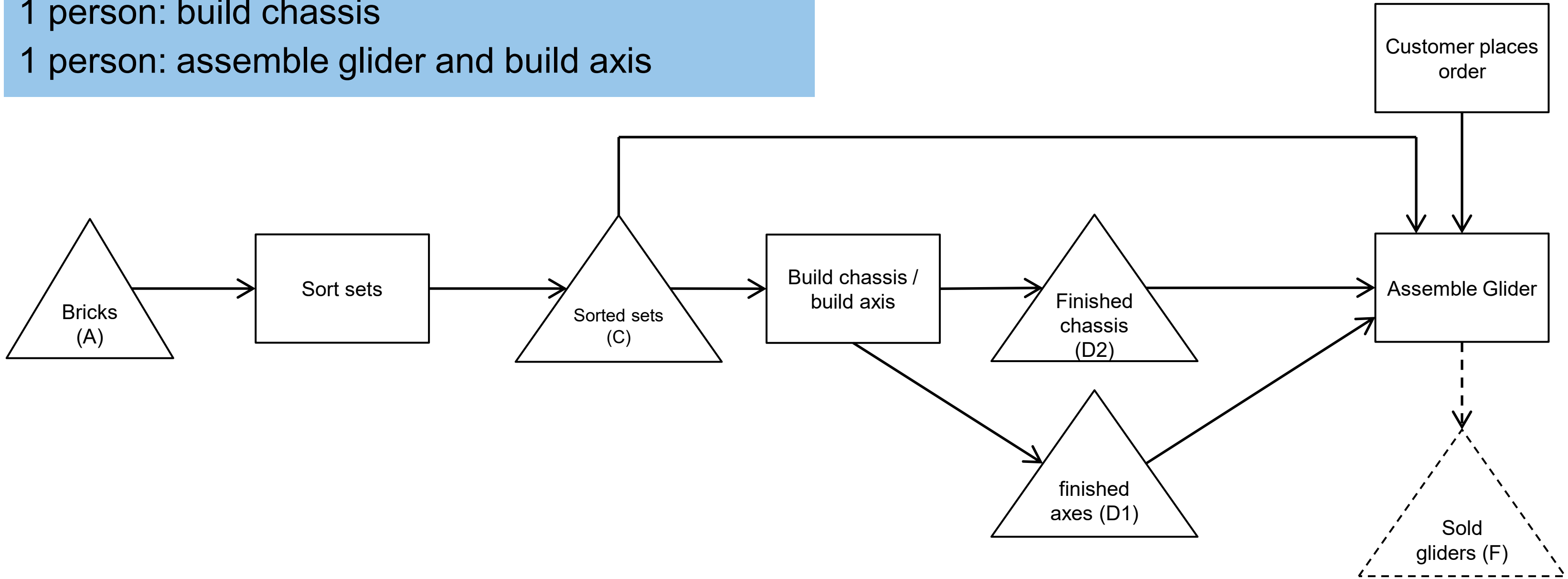
Five Concepts of Flow

- standardization
- takt time
- no rework
- just-in-time
- transparency/visual control

Iteration 3: Manufacturing Cells

Iteration 3: Manufacturing Cells

1 person: sort sets
1 person: build chassis
1 person: assemble glider and build axis



Sequence

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Taking inventory

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Taking inventory

3rd production run: 3 minutes production (1 minute corresponds to 1 week in real)

Taking inventory

4th production run: 3 minutes production (1 minute corresponds to 1 week in real)

Taking inventory

Timer (3 minutes)

Minute 1

Minute 2

Minute 3

Start!

please stop producing!

Debrief: Iteration 3

What’s next?

Central Concepts of Lean Management

Kaikaku

Kaizen

Muda (Waste)

- overproduction ahead of demand
- unnecessary transport of materials
- over-processing of parts
- inventories above the minimum
- unnecessary movements by employees during their work
- production of defective parts
- waiting

Poka-yoke

Bonus round #4

How lean can you get?

Bonus round – How lean can you get?

- Product redesign and standardization?
- Sales and operations planning?
- Postponement?
- Factory layout?
- Work cells and assembly line?